ITA0608-MACHINE LEARNING

LAB PROGRAMS

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EXP NO:3 Tree based ID3 Algorithm

AIM:

To implement and demonstrate the Tree based ID3 algorithm for finding the most specific hypothesis from the given training data

PROCEDURE:

1. Define the training dataset
2. Initialize the hypothesis with the most specific values (0).
3. Consider only the positive training examples
4. Update the hypothesis by replacing mismatched attributes with ?
5. Display the final hypothesis

PROGRAM:

```
# ID3 Decision Tree Algorithm
import pandas as pd
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier, export_text
# Training Dataset
data = {
    'Outlook': ['Sunny', 'Sunny', 'Overcast', 'Rain', 'Rain',
                'Rain', 'Overcast', 'Sunny', 'Sunny', 'Rain',
                'Sunny', 'Overcast', 'Overcast', 'Rain'],
    'Temperature': ['Hot', 'Hot', 'Hot', 'Mild', 'Cool',
                    'Cool', 'Cool', 'Mild', 'Cool', 'Mild',
                    'Mild', 'Mild', 'Hot', 'Mild'],
    'Humidity': ['High', 'High', 'High', 'High', 'Normal',
                 'Normal', 'Normal', 'High', 'Normal', 'Normal',
                 'Normal', 'High', 'Normal', 'High'],
    'Wind': ['Weak', 'Strong', 'Weak', 'Weak', 'Weak',
             'Strong', 'Strong', 'Weak', 'Weak', 'Weak',
             'Strong', 'Strong', 'Weak', 'Strong'],
    'Play': ['No', 'No', 'Yes', 'Yes', 'Yes',
             'No', 'Yes', 'No', 'Yes', 'Yes',
             'Yes', 'Yes', 'Yes', 'No']
}
df = pd.DataFrame(data)
# Encode categorical data
le = LabelEncoder()
X = df.drop('Play', axis=1)
y = le.fit_transform(df['Play'])
encoders = {}
for col in X.columns:
```

```python
for col in X.columns:
    encoders[col] = LabelEncoder()
    X[col] = encoders[col].fit_transform(X[col])
# Train ID3 Decision Tree
model = DecisionTreeClassifier(criterion='entropy')
model.fit(X, y)
# Print Decision Tree
print("Decision Tree:\n")
print(export_text(model, feature_names=list(X.columns)))

# New sample for prediction
new_sample = pd.DataFrame({
    'Outlook': ['Sunny'],
    'Temperature': ['Cool'],
    'Humidity': ['High'],
    'Wind': ['Strong']
})
for col in new_sample.columns:
    new_sample[col] = encoders[col].transform(new_sample[col])
prediction = model.predict(new_sample)
print("\nPrediction:")
print("Play =", le.inverse_transform(prediction)[0])
```

OUTPUT:

```
... Decision Tree:

|--- Outlook <= 0.50
|   |--- class: 1
|--- Outlook >  0.50
|   |--- Humidity <= 0.50
|   |   |--- Outlook <= 1.50
|   |   |   |--- Wind <= 0.50
|   |   |   |   |--- class: 0
|   |   |   |--- Wind >  0.50
|   |   |   |   |--- class: 1
|   |   |--- Outlook >  1.50
|   |   |   |--- class: 0
|   |--- Humidity >  0.50
|   |   |--- Wind <= 0.50
|   |   |   |--- Outlook <= 1.50
|   |   |   |   |--- class: 0
|   |   |   |--- Outlook >  1.50
|   |   |   |   |--- class: 1
|   |   |--- Wind >  0.50
|   |   |   |--- class: 1


Prediction:
Play = No
```

RESULT :

Thus ,the tree based ID3 algorithm was implemented successfully ,and the most specific hypothesis was obtained .